pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

ROC Curves and Choosing a Threshold

WEEK 2 · DAY 3

Created by **Jeff Lenamon**

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Where day 7 left off

Day 7: threshold 0.5

324 of 1,327

defaults caught, 6,000 validation rows

Recall 0.2442, precision 0.7248. Nothing about 0.5 says it is the place to stop.

Real data

UCI card file: accounts in Taiwan, 2005. Not US credit today.

People's attributes

SEX, EDUCATION, MARRIAGE, AGE: never features.

60 / 20 / 20

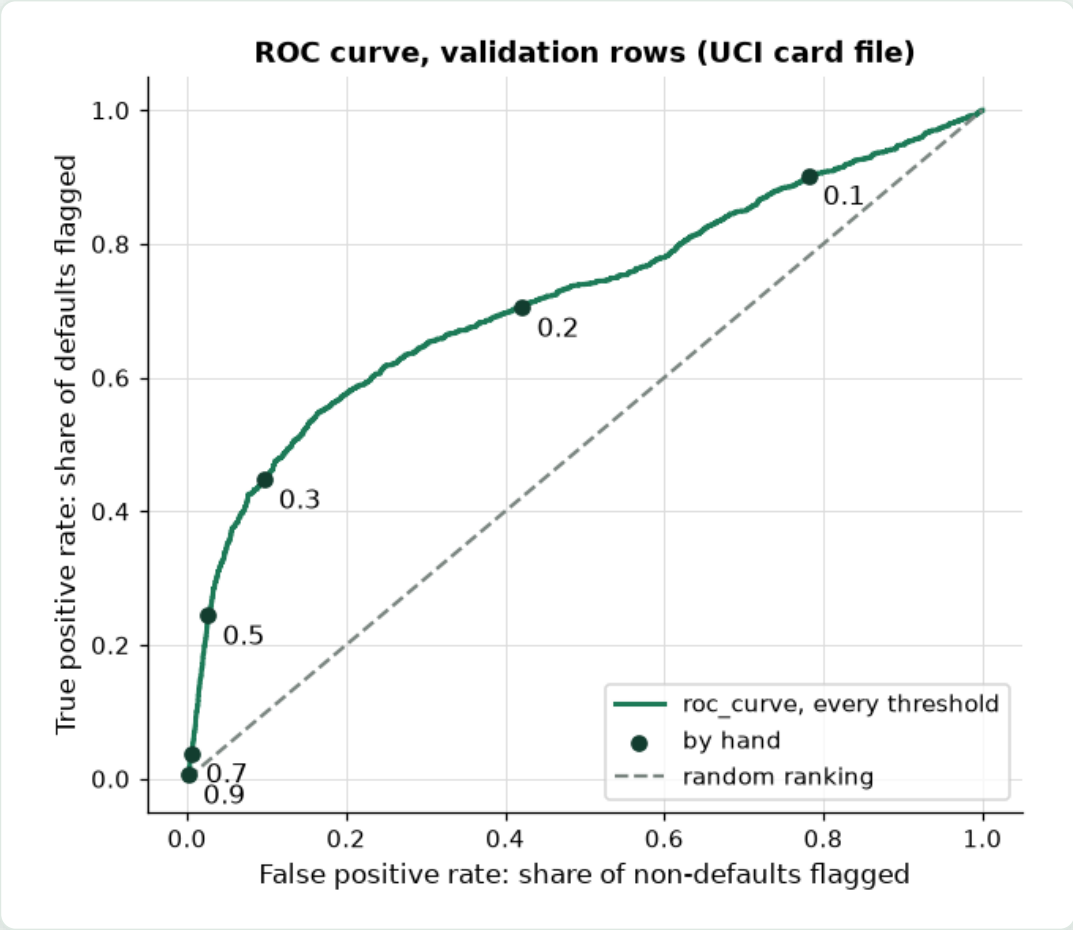
Training fits, validation chooses, test reports once (day 10).

Names

Model probability of default in this file; flagged at threshold t.

Every threshold and target today is a teaching convention, not what a lender does or should do.

Every threshold at once: the ROC curve

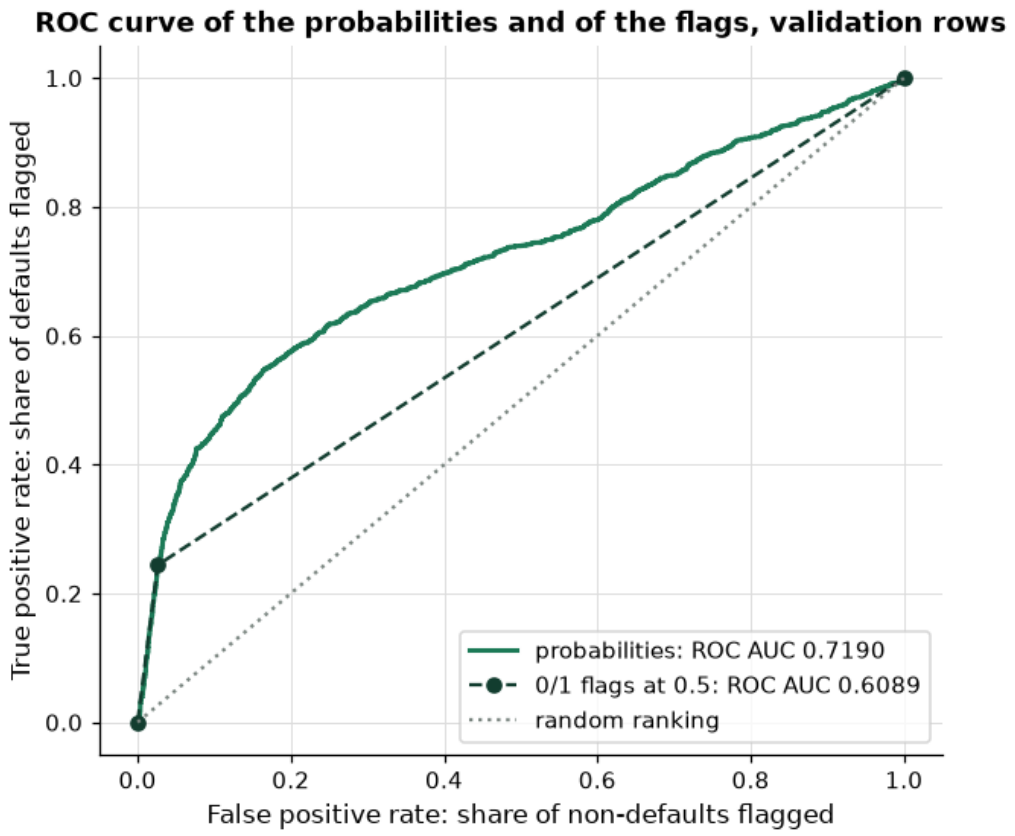


Rate	Formula	Out of
True positive (recall)	TP / (TP + FN)	1,327 defaults
False positive	FP / (FP + TN)	4,673 others

Threshold	Flagged	TPR	FPR
0.5	447	0.2442	0.0263
0.3	1,047	0.4476	0.0969
0.2	2,905	0.7046	0.4216
0.1	4,848	0.9020	0.7813

Up the curve is a lower threshold. The diagonal is a score with no information.

AUC as a ranking measure



0.7190

ROC AUC, probabilities

0.6089

the 0/1 flags at 0.5

One account that defaulted, one that did not: AUC is the chance the model ranks the defaulted one higher. Here, 71.90 percent of 6,201,071 pairs.

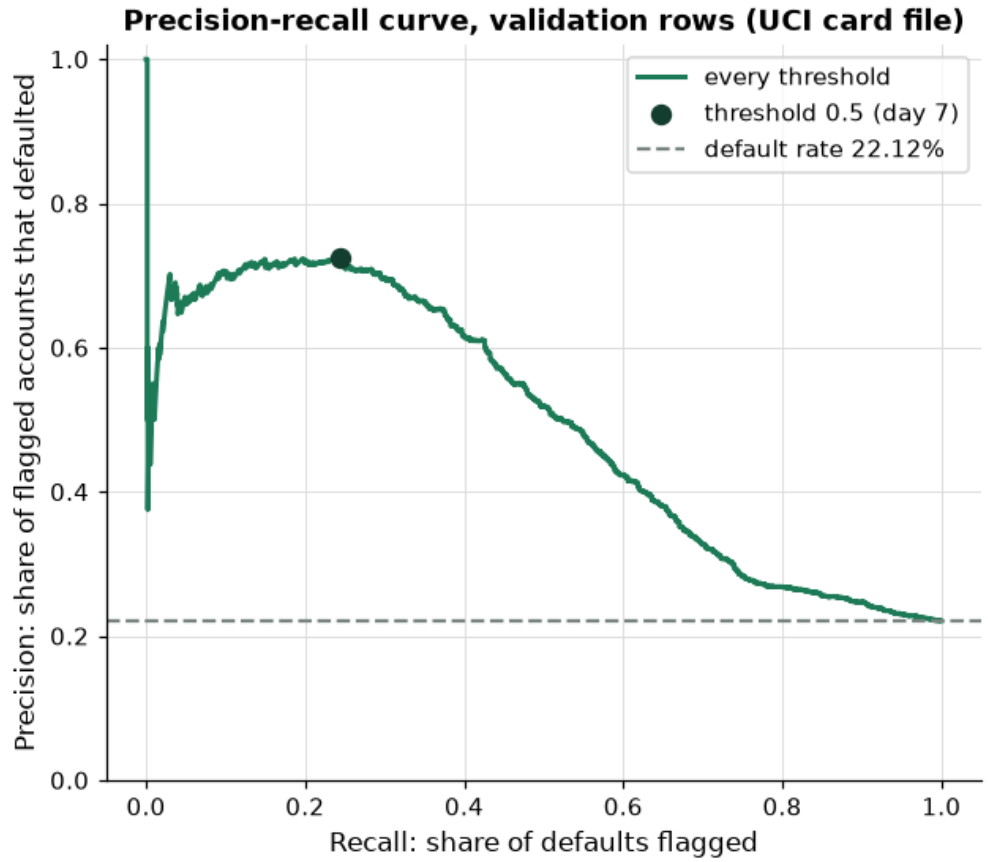
What AUC does not say

Not an accuracy, not a default rate

Nothing about any one threshold

Not whether a 0.3 means three in ten: squared, 0.7190

The precision-recall curve



0.4946

average precision

22.12%

the floor: the default rate

Precision divides by the flagged accounts: of those flagged, how many defaulted? The false positive rate divides by every non-default.

Average precision is a step sum: each rise in recall times the precision there. Not a trapezoid.

Choosing a threshold on validation rows

The course's rule: mlkit.threshold_for_recall

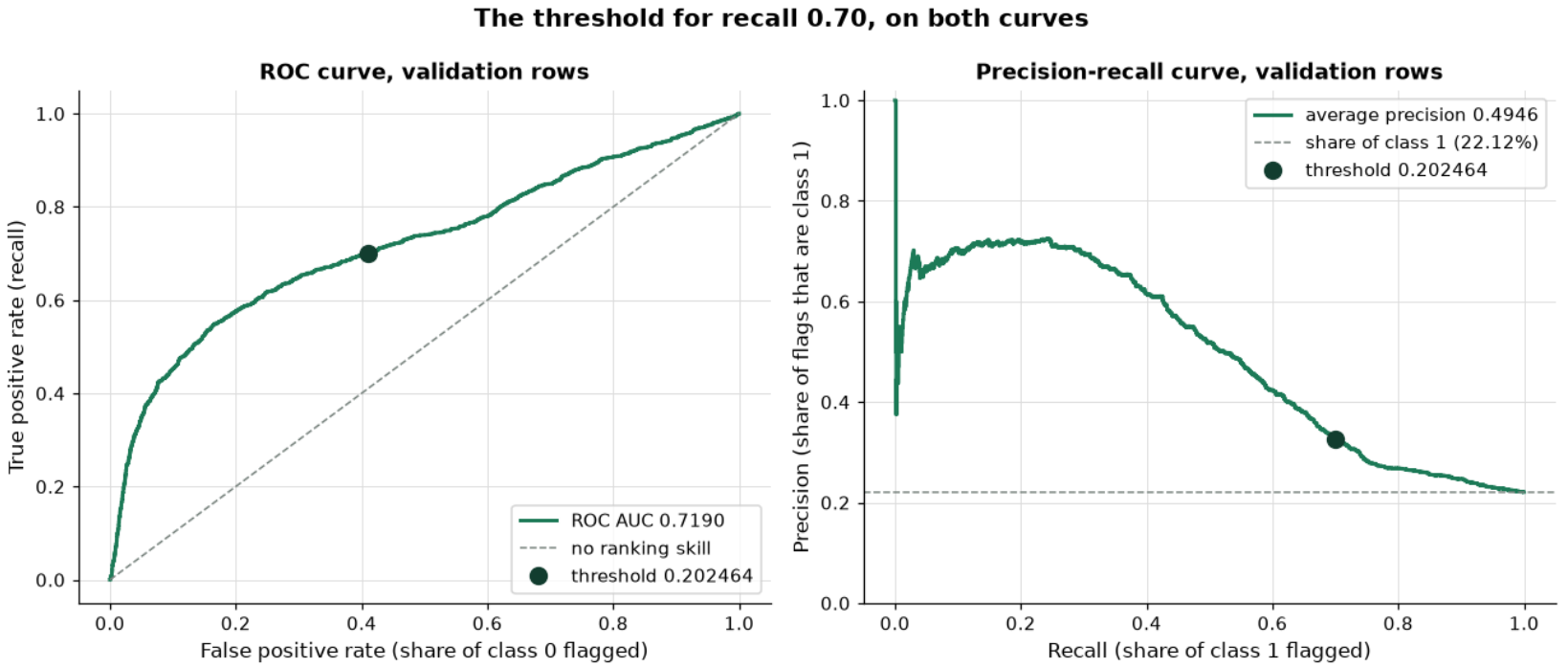
If the desk asked for recall of 70 percent: the highest threshold with validation recall at least 0.70.

Rule	Threshold	Flagged	Recall	Precision	F1
Day 7: 0.5	0.5000	447	0.2442	0.7248	0.3653
Round 0.2	0.2000	2,905	0.7046	0.3219	0.4419
Recall at least 0.70	0.2025	2,843	0.7001	0.3268	0.4456
Youden's J	0.2475	1,507	0.5494	0.4837	0.5145

Youden's J: where TPR minus FPR is largest (0.3829). Accuracy of flagging nothing: 0.7788; never used to choose.

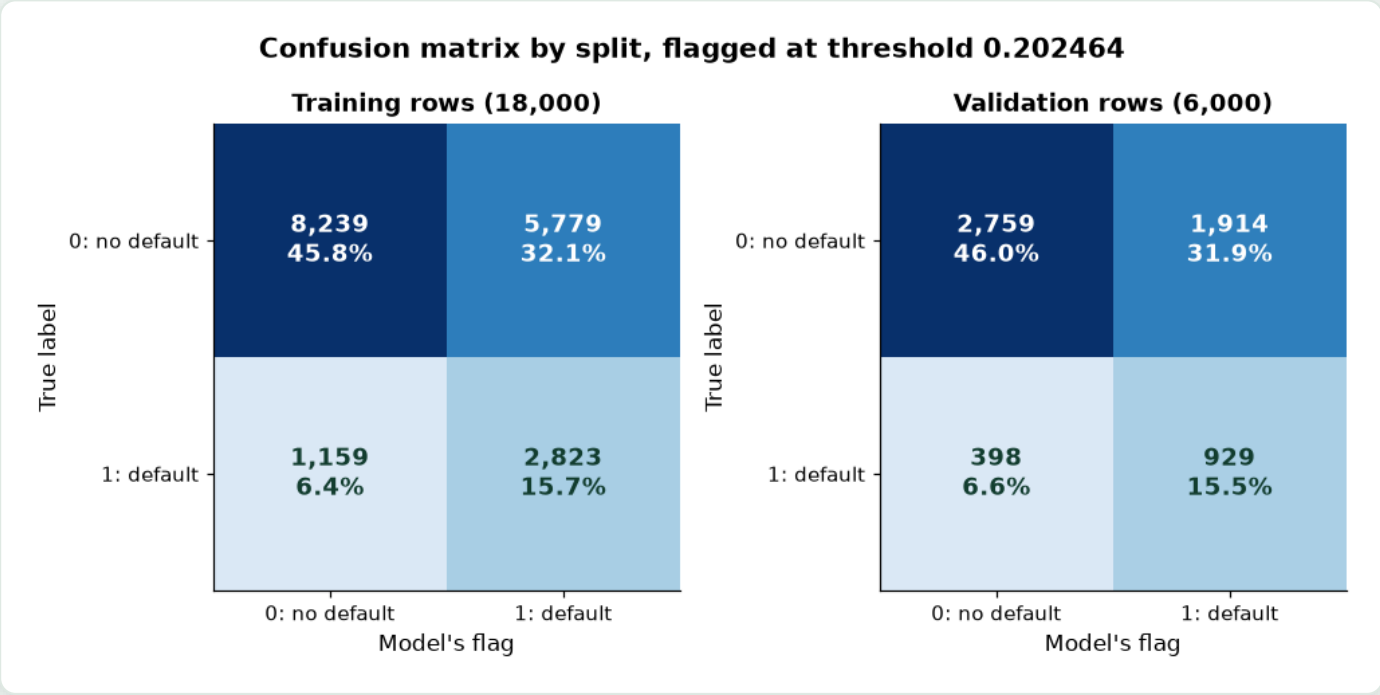
A teaching convention, not a lending standard. No threshold here is what a lender does or should do.

The chosen threshold on both curves



One threshold, one point on each curve: recall 0.7001, precision 0.3268. About one flagged account in three defaulted.

The chosen threshold, training and validation rows

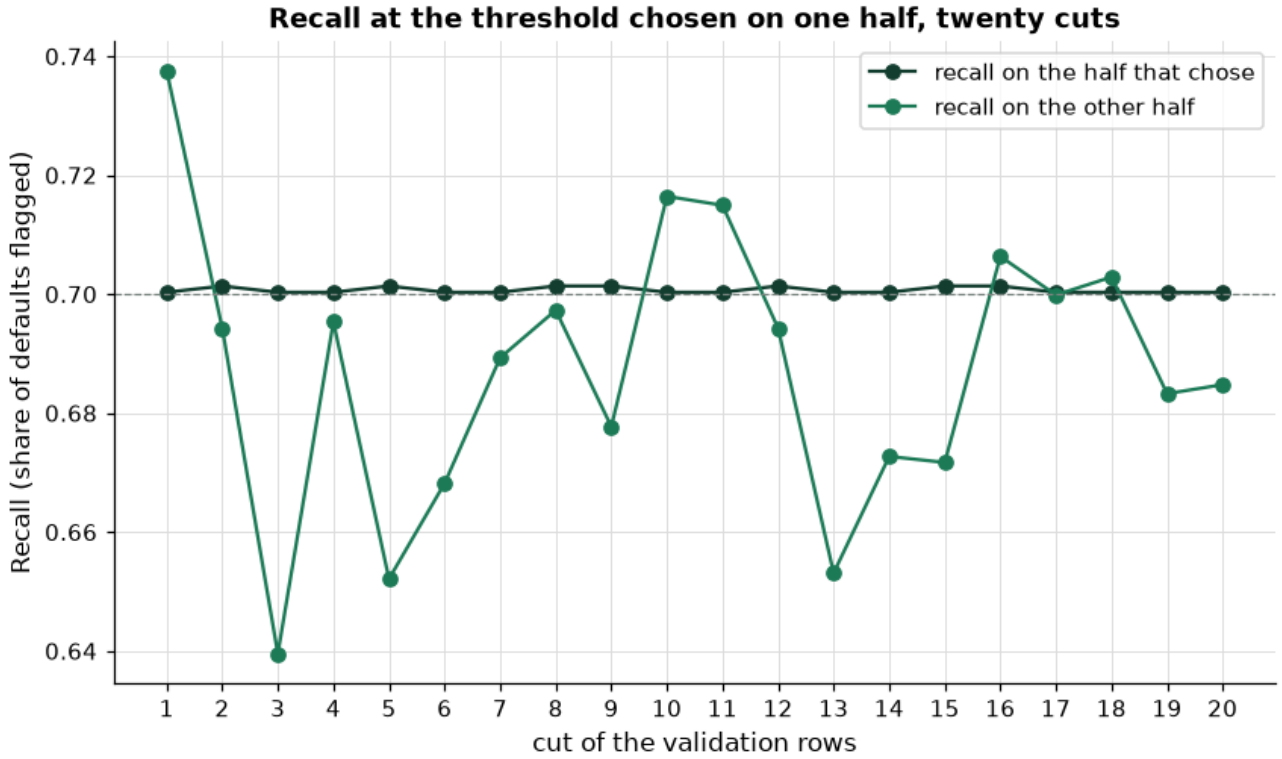


Rows	Recall	Precision	F1
training	0.7089	0.3282	0.4487
validation	0.7001	0.3268	0.4456

Read the panels as shares: the splits differ in size. The two rows are close.

The validation rows chose this threshold, so their recall is not the number to report.

Why the test rows are not used to choose



0.7003 to 0.7014

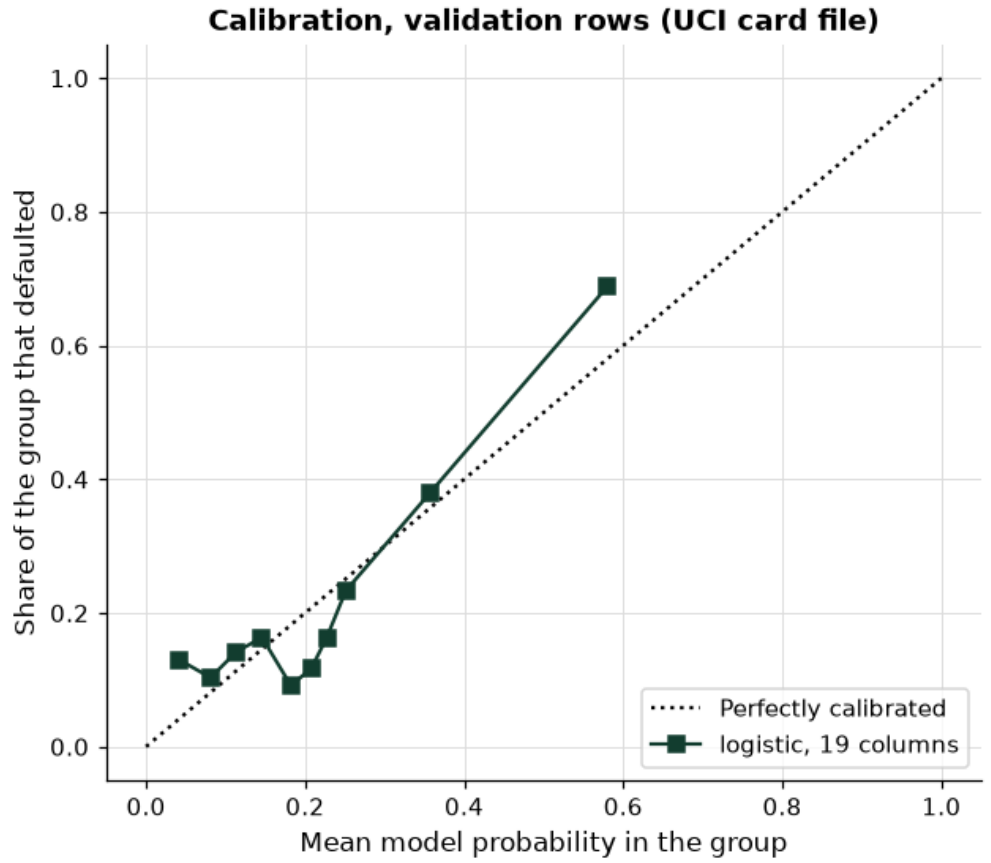
the half that chose

0.6395 to 0.7376

the other half: below 0.70 in 15 of 20

Chosen on the test rows, the reported recall would be 0.70 by construction. Day 10 opens them once.

Calibration: does a 0.3 mean three in ten?



Group	Mean probability	Defaulted
Lowest	0.0416	13.00%
Nearest 0.3	0.2512	23.33%
Top	0.5800	68.83%

Ten groups of about 600 accounts, sorted by probability. On the diagonal, a probability means what it says.

It ranks better than chance; its probabilities are not uniformly reliable as numbers. Recalibration: Course 5.

Excel side by side

Excel side by side (tested in Excel)

`=COUNTIFS(A2:A6001,1,B2:B6001,">="&D2)  (row 2)`	TP 324 at 0.5
`=F2/COUNTIF(A2:A6001,1)`	recall 0.2442
`=SORT(UNIQUE($B$2:$B$6001),,-1)  (M3)`	5,882 thresholds
`=COUNTIFS($A$2:$A$6001,1,$B$2:$B$6001,">="&M3#)/COUNTIF($A$2:$A$6001,1)  (O3)`	TPR, spilled
`=SUMPRODUCT((N3:N5884-N2:N5883),(O3:O5884+O2:O5883)/2)`	AUC 0.7190
`=MAXIFS(M3#,O3#,">=0.7")`	rule 0.2025
`=INDEX(M3#,MATCH(MAX(O3#-N3#),O3#-N3#,0))`	Youden 0.2475
`=SUMPRODUCT((O3:O5884-O2:O5883),P3:P5884)`	AP 0.4946

Excel has no ROC or AUC function: the probabilities come from Python; every step after them is a formula.

Save the day: the history of the choices

```
git log --oneline -- results.csv
  Experiment log: Youden's J threshold beside it, for comparison
  Experiment log: the threshold for recall 0.70, chosen on validation rows
  Experiment log: the logistic model at day 7's threshold 0.5
```

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
logistic_scaled at fixed 0.5,validation,0.7190,0.4946,0.5000,0.8123,0.7248,0.2442,0.3653
logistic_scaled at recall at least 0.70,validation,0.7190,0.4946,0.2025,0.6147,0.3268,0.7001,0.4456
logistic_scaled at Youden J,validation,0.7190,0.4946,0.2475,0.7707,0.4837,0.5494,0.5145
```

One row per choice, one commit per row. The two dashes keep only the commits that changed results.csv.

Why the threshold is what it is: the log shows the order of the choices and the commit behind each.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the threshold is chosen on the test rows, with a comment that says the model never saw them.

Code that runs is not the same as code that is right.

What you can do now

- The ROC curve: every threshold at once, each point a confusion matrix
- AUC 0.7190: pairs ranked the right way, nothing about one threshold
- Average precision 0.4946: a step sum over the precision-recall curve
- Recall at least 0.70 on validation rows: threshold 0.2025, a convention
- The rows that choose cannot report; calibration checks the numbers

Week 2, Day 4 Decision trees: a model made of questions, on the same validation rows